

Real Time flood forecasting and Flood Management – Experience from **National Hydrology Project (NHP)**

Dr. Devadasan Gnanasundar

Senior Joint Commissioner

National Hydrology Project

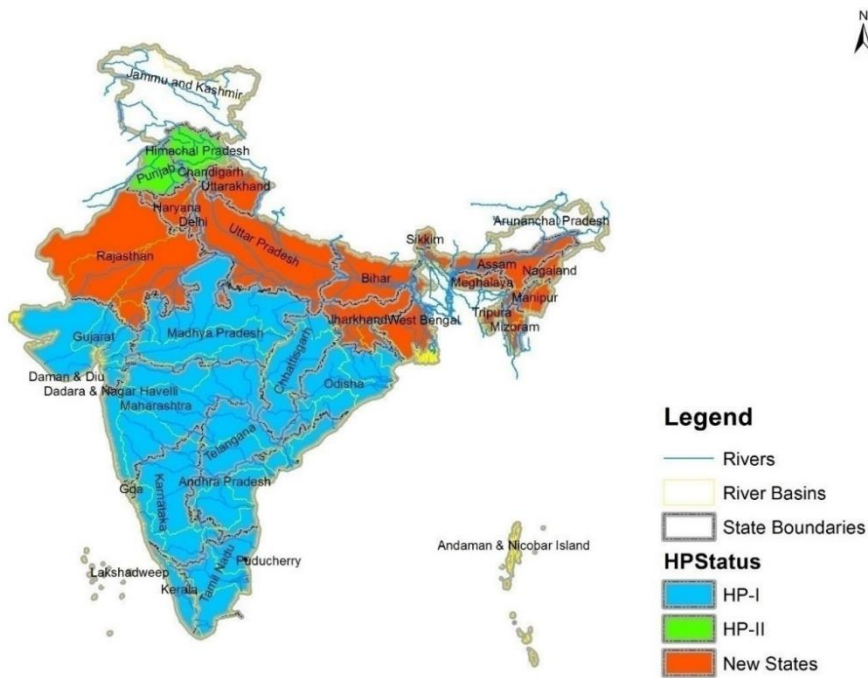
Ministry of Jal Shakti

Government of India

National Hydrology Project (NHP) - National Perspective on Flood forecast and Flood Management

Not the intensity but the location of RTDAS sites plays a crucial role in Flood forecast and management.

States under NHP



States under NHP (HP-I + HP-II + New States)

Agencies under NHP

Implementing Agencies	HP - I (1995-2003)	HP - II (2006-2014)	NHP (2016 – 2024)
States	9	19	34
UTs	-	2	2
Central Agencies	6	6	9
RBOs	-	1	3
Total	15	28	48

NHP: Major Highlights

- Pan-India coverage
- Web-based National Database – India WRIS
- Focus on Real time Data Acquisition
- Development of analytical tools
- River basin based water accounting
- Advanced Flood forecasting system
- Capacity building of institutions

RTDAS stations (AWLR/AWS) to be installed and commissioned in the country by March 2022.

- 4750 nos

Real Time flood forecasting and Flood Management – Experience from NHP

Flood Prone Area (India)

Major Flood Prone States

- Assam, Bihar, West Bengal, UP, Uttarakhand, Orissa.

Major Flood Prone Basins

- Ganga, Brahmaputra & Mahanadi

Ministry bulletin – flood forecasting.

Major rivers (Travel time >24 hours)

- a. Forecasts formulated based on 0800 hr/ 0900 hr
- b. water level data and forecast issued once in a day at 1000 hr with advance warning time from 24 hr to 36 hr

Medium rivers (Travel time 12-24 hours)

- a. Forecasts formulated based on 0600 hr & 1800 hr
- b. water level data and forecast issued twice in a day at 0700 hr & 1900 hr with advance warning time from 12 hr to 24 hr

Flashy rivers (Travel time < 12 hours)

- a. Forecasts formulated based on any hr
- b. water level data and forecast issued multiple times (more than twice) in a day with advance warning time less than 12 hr

Flood Control and Flood Management

- a. Floods are natural phenomena.
- b. Complete immunity is not feasible, their impacts can be minimized where loss of life is nil.

Flood Management with combination of structural and non-structural measures is the most feasible solution.

Structural Measures	Non-Structural Measures
Dams and Reservoirs	Flood Forecasting and Warning
Embankments, Flood Walls, Sea Walls	Flood Plain Zoning
Natural Detention Basins	Flood Proofing
Channel Improvement	Disaster Preparedness and Response Planning
Drainage Improvement	Disaster Relief
Diversion of Flood Water	

Flood Forecasting and Warning : Non-Structural Measures.

Key initiatives/activities under NHP – Addressing Flood forecasting and Flood management in India

○ Institutional

- Modification of Hydro-met Data Dissemination Policy – Modified in Nov. 2018
- Establishment of National Water Informatics Centre (NWIC)- Established in Mar. 2018.
 - NWIC is now the custodian of water resources information/WIMS./WRIS
- Capacity building of Implementing Agencies and Infrastructure strengthening
 - in all NHP states/agencies
- Establishment of data centres – initiated in all NHP states/agencies.

○ Data Acquisition Modernisation

- Densification of surface RTDAS network- 4750 nos
- Installation of SCADA on selected projects-
- Development of high resolution Digital Elevation Models – About 70,000 sq. km covered (60,000 sq.km in flood prone Ganga basin & 10,000 sq.km in remaining regions)

National Hydrology Project – Concept/Strategy

Modernizing monitoring

RTDAS

(Real Time Data Acquisition System)



Establish monitoring networks in states, with a focus on deploying new sensors, data storage, capturing high frequency data with telemetry. (AWLR/AWS)

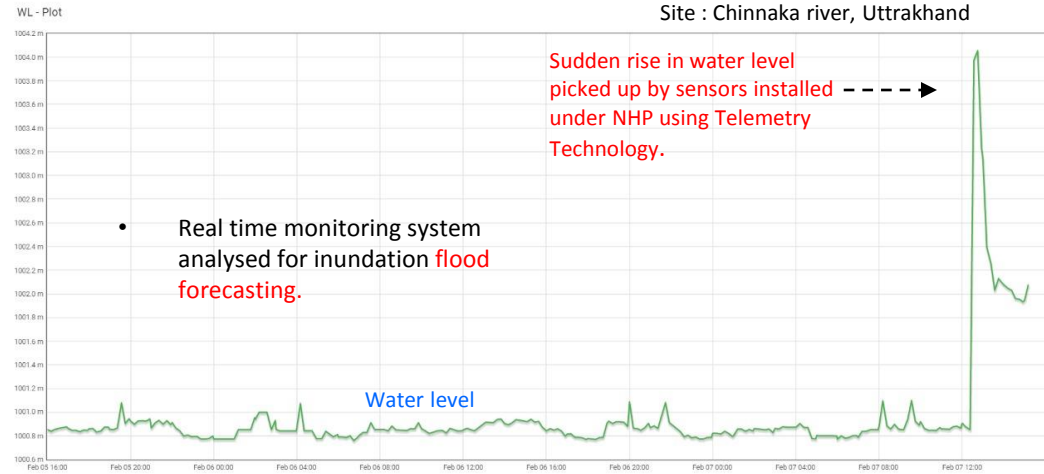
For comprehensive, automated, real-time monitoring of surface.

Outcome

1. Strong data base on water resources in states/central agencies.
2. Comprehensive data management systems nation wide. (India WRIS/State WRIS/WIMS)

Enhancing analytical tools

Demonstrate tools for water resources assessment, hydrologic & flood inundation forecasting, water infrastructure operations, groundwater modelling, river basin/aquifer management plan.



Transforming knowledge access

Advances in cloud computing internet, mobile devices, social media to modernize access/ visualization of customized water information to different stakeholders.

Modernizing institutions

Investment in people/institutions. Develop centres of Excellence, collaboration with Research Institutes. Modernise/ streamline workflow in states.

Final Output (Water Resources Management)

River Basin Management

Outcome & Impact of major studies

Flood early warning system

- Inundation forecast with increased lead time to reduce flood deaths
- Base work for floodplain regulatory management to minimise flood loss

DEM/DTM and geodatabase

- Flood inundation modelling & evacuation planning
- Water resources project planning

Extended hydrological prediction

- Better reservoir management to cope with uncertainty
- Better demand and supply management

National Water framework and web enabled hydrological model

- Online tool for Water availability assessment
- Resource optimisation through scenario modelling

Optimised multi-reservoir management

- Improved water use efficiency through optimised river basin management
- Increased net benefits at basin scale across the states

Reservoir sedimentation survey and analysis

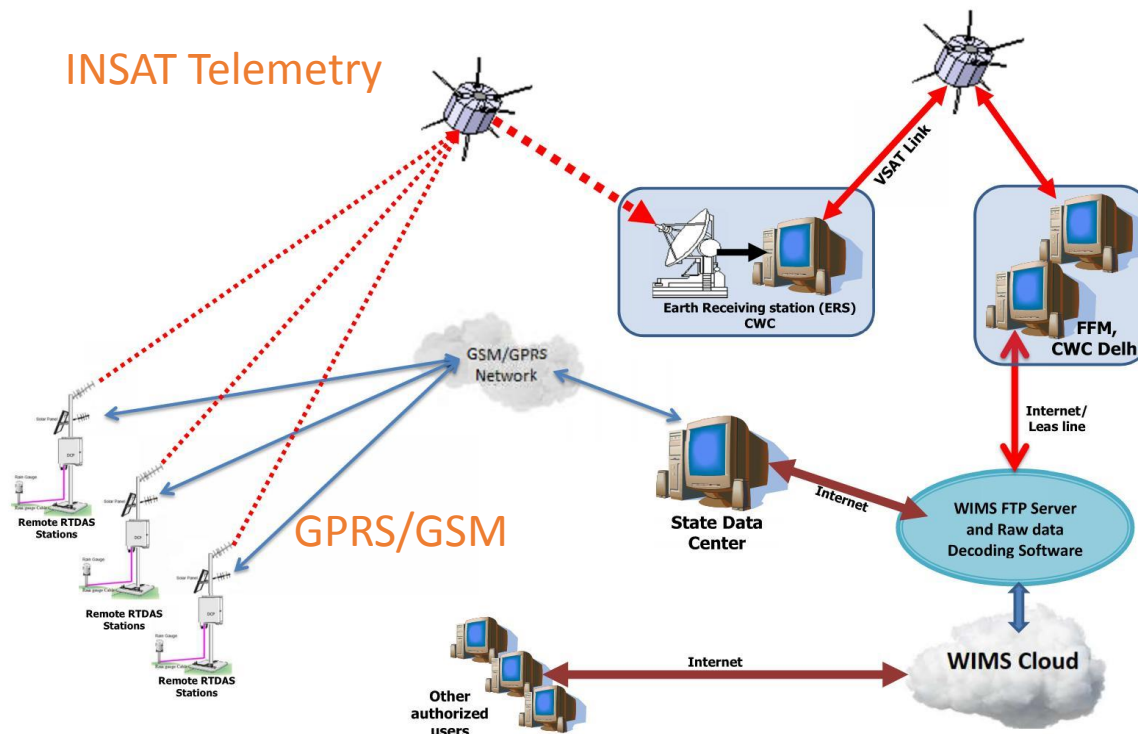
- More accurate estimation of stored water
- Better reservoir operation, minimising loss and maximising utilisation

Real Time flood forecast and flood management (sequence)

- Hydromet network monitoring committee
- Procurement, Installation and commissioning
- Data Transmission to State Data Centre/WIMS

6 to 9 months – from Award

- **Data acquired every 60 minutes.**
- **If required, frequency can be increased to every 15 minutes.**



Transmission of Data from RTDAS station to central portal/WIMS

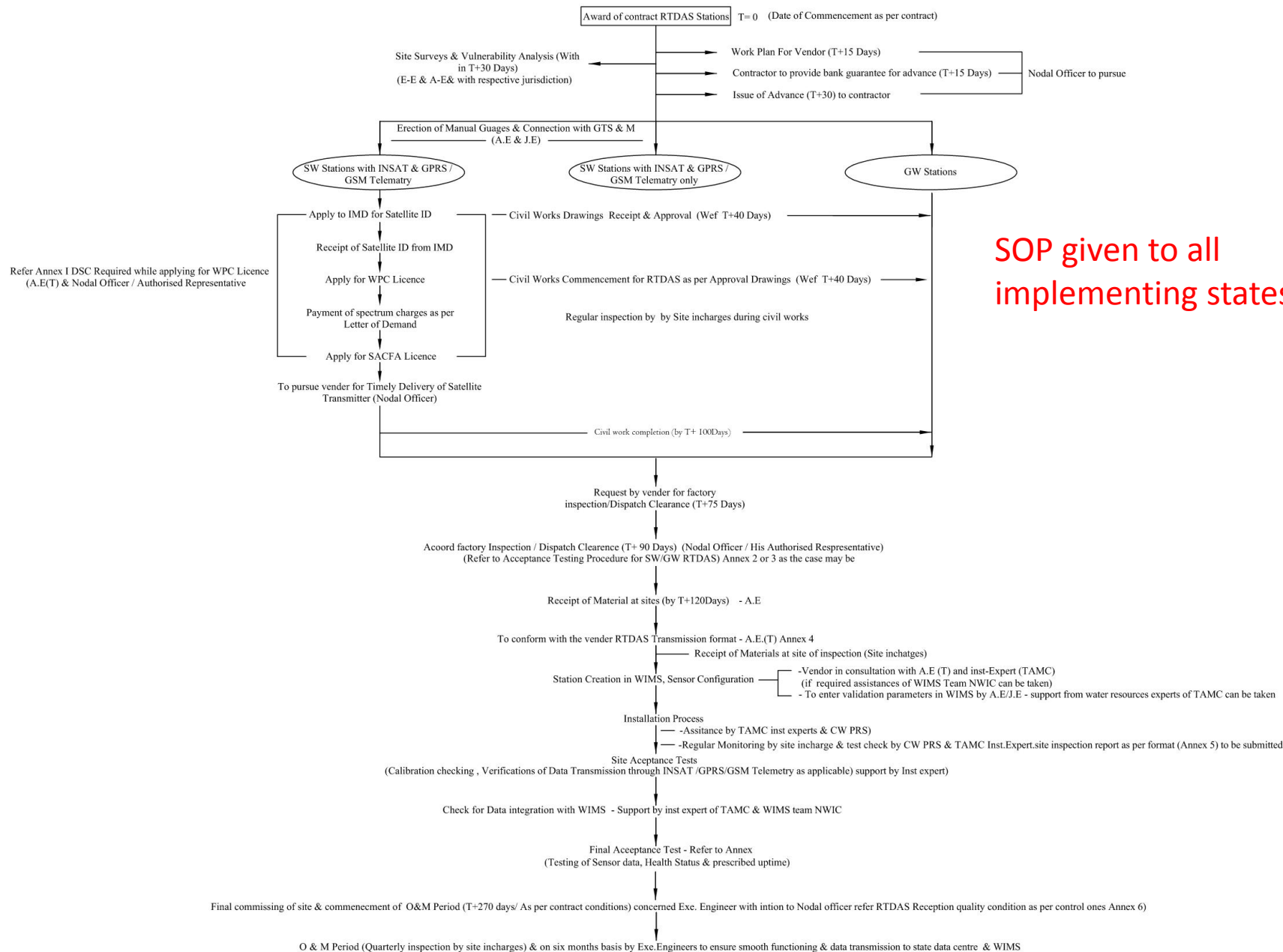
Station Configuration
From sensor to Data
Logger
To FTP server
WIMS platform.

- Flood Forecasting by Experts
- Advisories to states/information to public/departments.

INSAT Telemetry – WPC license

SOP - RTDAS installation, Commissioning and Data Transmission to state data centres/WIMS

Standard operating procedure RTDAS installation Commissioning Including Data Transmission to state data centres & WIMS



RTDAS – in Telangana State

RTDAS station in NHP states

- AWLR – Sensor

Visit or Inspection to site is much vital.

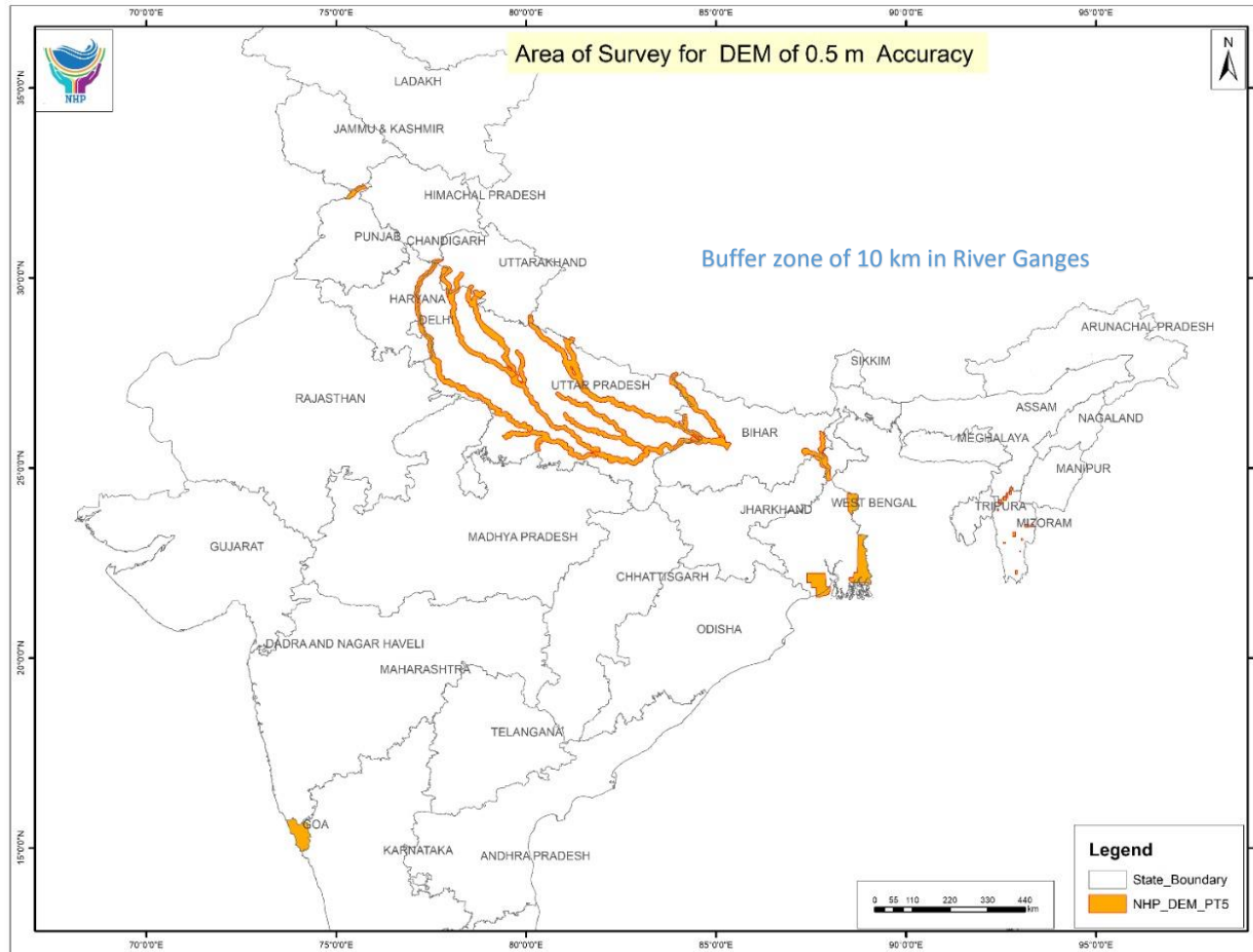
Any delay in Data Transmission or non-Transmission of data to be attended in 24 hours.

- Data Logger



DEM - To demarcate zones/regions inundated by flood-

Requirement of spatial data - DEM of 0.3 m resolution



Sol in carrying DEM survey under NHP

Agency can approach Sol for DEM data.

Web enabled advanced flood forecast study by NRSC under NHP - Objectives of the Study



To develop medium-range (**2 – 3 days advance**) flood early warning models for Godavari and Tapi Rivers using space based inputs

To develop spatial flood inundation simulation models using high resolution Digital Elevation Models



To develop web enabled real time spatial flood early warning system.

To develop **workflow mechanism for issuing flood advisory** to the concerned DMS authorities during the flood event.



Scientific assessment of flood prone area in both the basins under different flood return periods

Flood Early Warning : Approaches and Space Inputs

Parameter	Conventional Technique	Modelling Technique
Range	Short range forecast	Medium to long range
Approach	Gauge to gauge correlation	Distributed modelling
Accuracy	Less (if intermediate catchment plays major role)	More accurate
Spatial flood simulation	Not possible	Possible
Data requirement	Minimum field data required	More data on topography and meteorology required

Data Requirements

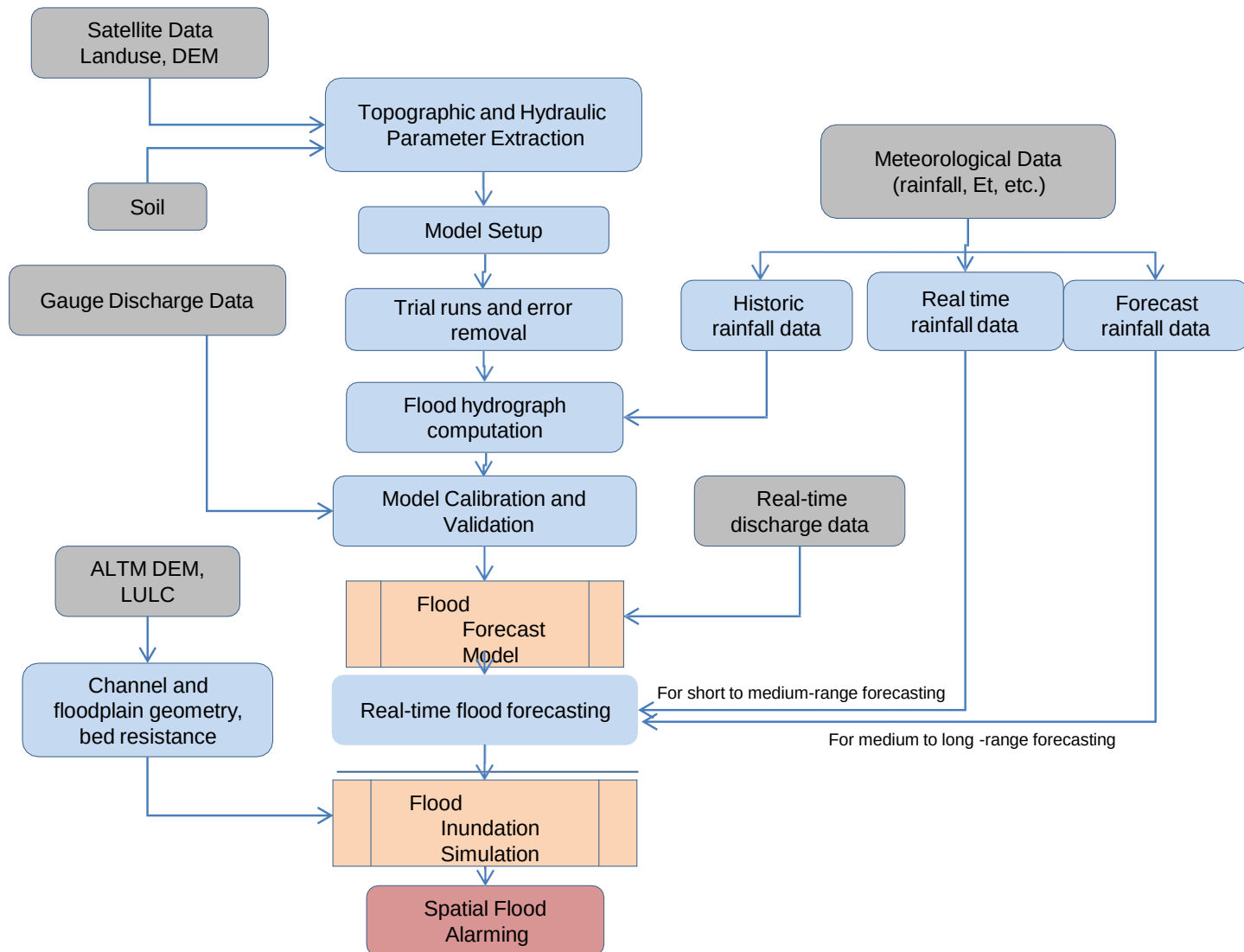
Static Data

Landuse/landcover, Soils ,
DEM, ALTM-DTM, Rating
Curves, etc.

Dynamic Data

- Real-time discharge data
- Daily Rainfall Data in
real- time, ET data
- Historic hydro-met data
for model cal &
validation.
- Reservoir operations data

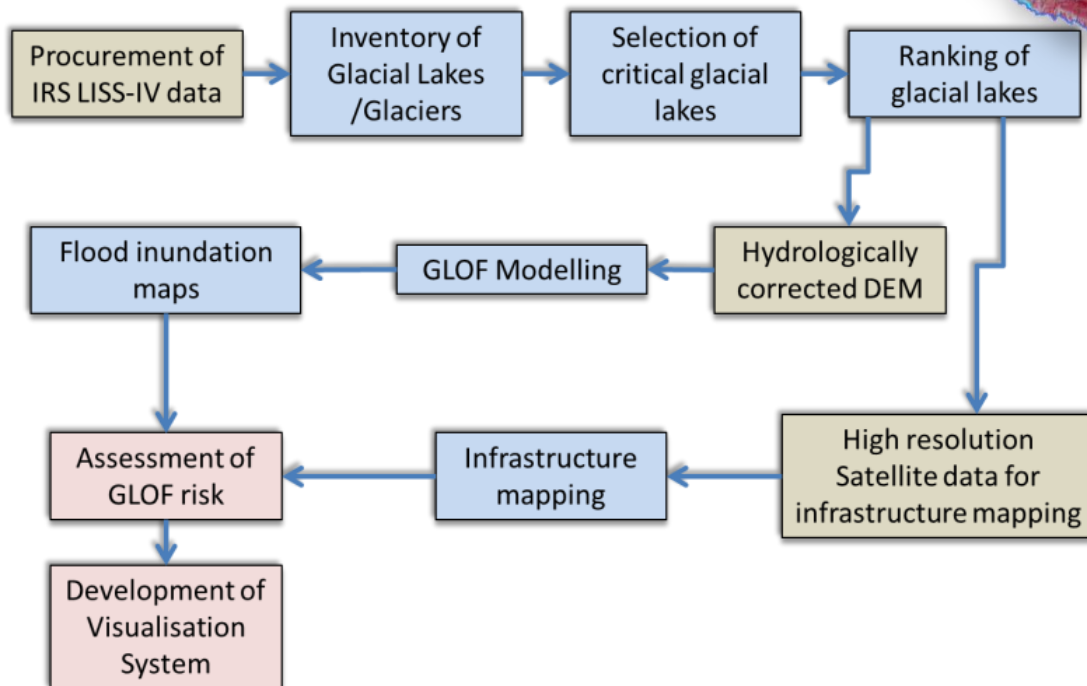
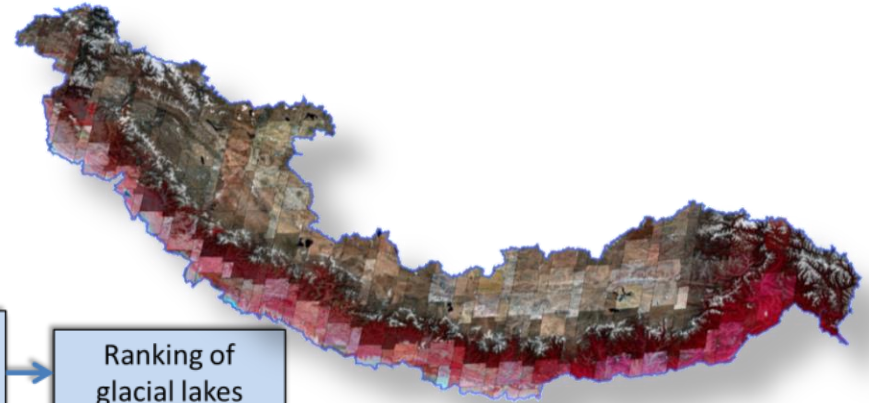
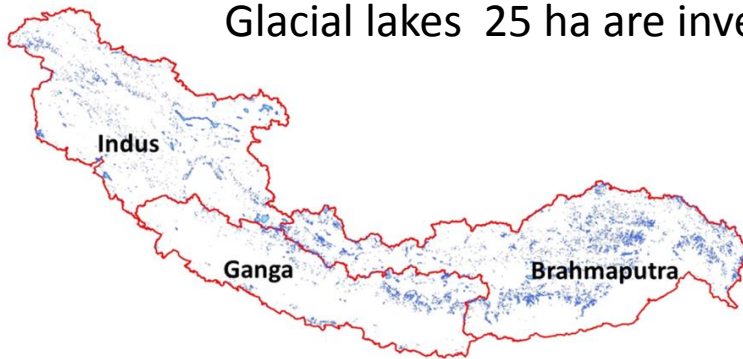
The Broad Methodology (using spatial data)



Glacial Lake Outburst Flood Risk Assessment in Catchment Area of Indian Himalayan River Basins

Glacial lakes 25 ha are inventoried.

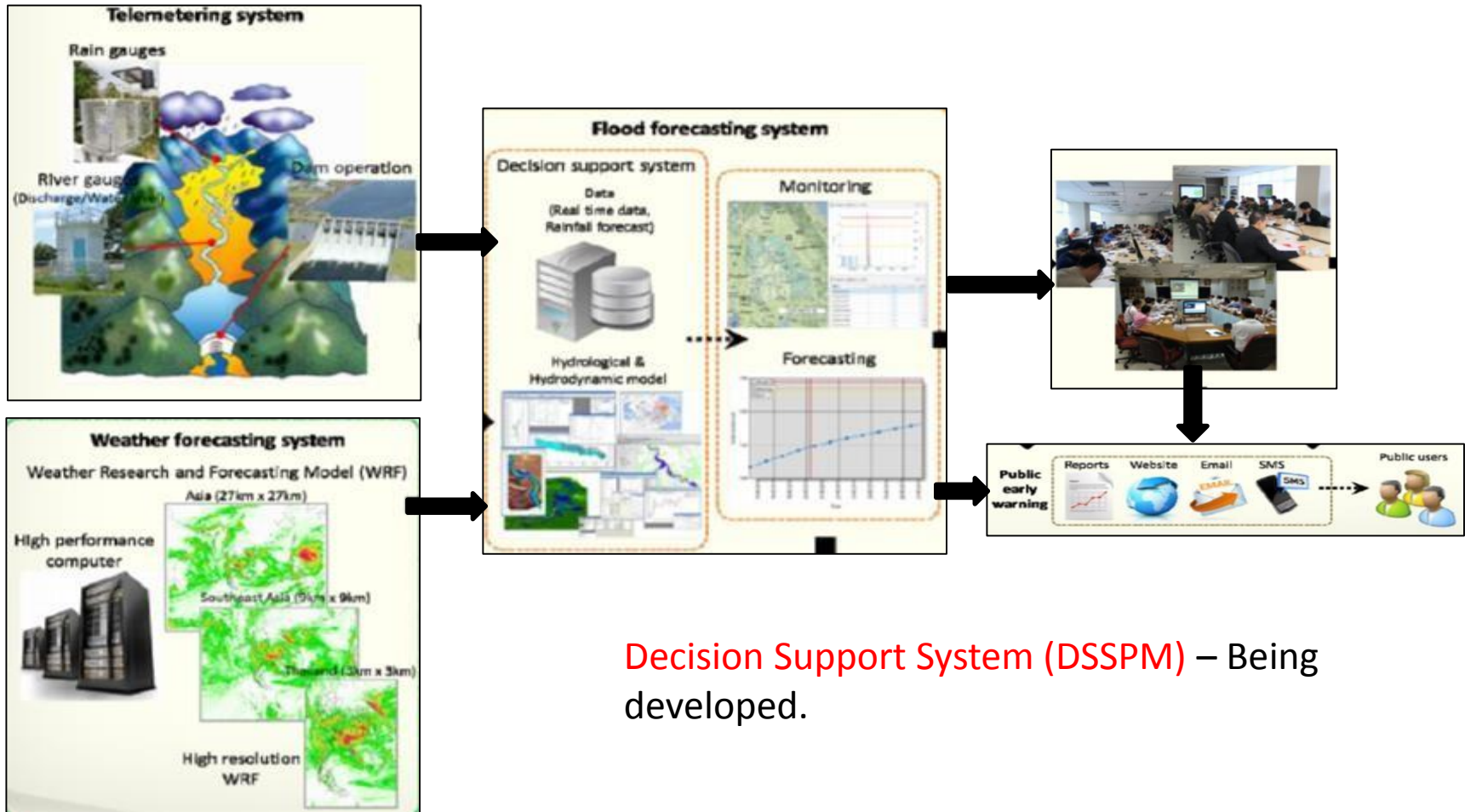
- RS-2 LISS-IV data (450 sub-scenes) covering entire study area



Critical lakes are identified.

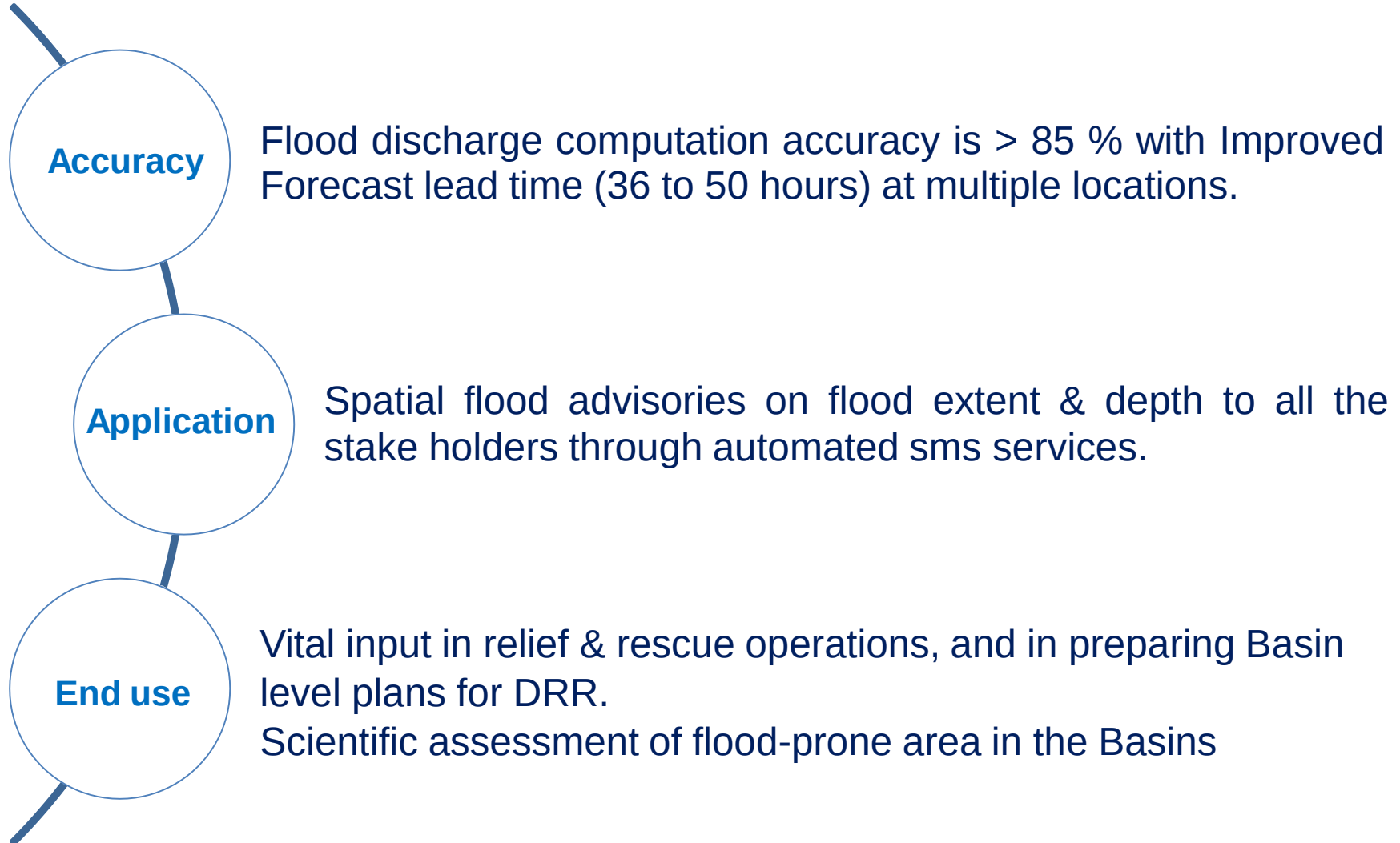
Impact of glacial lake bust is quantified by GLOF modelling,

Conceptual Spatial Flood Early Warning System



Decision Support System (DSSPM) – Being developed.

Major Highlights & End use



Real Time flood forecasting and Flood Management – Experience from NHP

Highlights

Development of fully automated web enabled real time operational flood early warning system by integrating reservoir operations data

Support from CWC/states for real time hydrological data transfer (discharge, gauge, reservoir operation data etc) for running the model in real time

Support from IMD for real time and forecasted meteorological data transfer (rainfall etc) for running the model in real time

Scientific assessment of flood-prone area/ Meeting with Stakeholders/ Transfer of technology and models

Way forward...

- Data Transmission to central portal/WIMS.
- Delay in installation and commissioning of RTDAS stations (AWLR)
- Transmitters - issue/make
- O & M – under NHP O & M constitutes for 5 years from commissioning of RTDAS stations.
- Experience in flood forecasting from different states are shared – further enhances the flood forecast model developed.
- [Handbook for flood protection, anti-erosion and river training works by Central Water Commission \(2012\)](#)
- - aims to provide necessary guidance to the field engineers in the state and central for design, appraisal, construction and monitoring of the flood management works to meet new challenges in the flood management in India.

- Thanks